

Air Pollution Prediction Using Sensor Data

Problem Statement

1 A city wants to predict **pollution spikes** by identifying locations with **low air pressure and high temperature**. Avg tem should be greater than 36 and pressure should be less than 1000

2 An IoT network provider detects **fake sensors** sending **duplicate temperature readings** every second.

3. An IoT analytics company needs to identify **which sensors send the most data per day**.

4. A **climate monitoring agency** wants to fetch **the most recent temperature** recorded by each sensor.

5.A company wants to **track all sensors manufactured by "XYZ Inc."** for warranty claims.

6. A **facility management company** wants to **replace sensors installed before 2024..**

7. An **energy company** wants to identify **sensors where the temperature jumped by more than 5°C between two consecutive readings**.

8.A company wants to find **sensors where metadata is missing** (e.g., no manufacturer info).

9. A meteorology team needs to **detect locations with large variations in pressure with value greater than 10**.

10. A **weather monitoring agency** needs to find sensors where the **temperature fluctuates by more than 5°C in the last 5 readings**.

11. A **smart home automation company** wants to retrieve sensor data **only for specific manufacturers (IoT Corp, SensorTech)** and specific **models (X100, Y200)** for performance evaluation.

12. A **network operations team** wants to identify IoT sensors that have been **inactive** (not transmitting data) for **more than 30 days**. The **last active date** is stored inside a **STRUCT** data type.